

Logic in Computer Science, Prolog 2

BITS Pilani- K K Birla Goa Campus

Total: 8 Marks

Weightage: 4%

Course No: *CS F214*

Time : 40 Minutes

November 02, 2023

Note:

- Store your file using your ID number. If it is not followed, you will get negative marks. For example if your ID no is 2014A7PS1234G then your file name should be 2014A7PS1234G.pl
 - Your file should contain four predicates (mysubset4/5, mylisttonum/2, myfib/2, mymerge/3).
 - You have to provide definitions of all the auxiliary predicates used and you are not allowed to use any libraries.
 - There should be no compilation error or warning messages when you run your prolog program. Any error or warning message, your program will not be evaluated and you will get negative marks.
 - The test cases might have variables as well not only just ground queries. For example, one test case for mytwice may be mytwice([a,b,c],Y).
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1. Write a prolog program with the following predicates

- (a) mysubset4(X1,X2,X3,X4,S) returns true iff X1 is subset of S, X2 is a subset of S, X3 is a subset of S, X4 is a subset of S. You can use the in-built predicate subset(U,V) (here U is a subset of V) to define mysubset4. (2 Marks)

Query: mysubset4(X,[1,2,3]). **Answer:** true

Query: mysubset4([1,2],[2,3],[],[4],[1,2,3,4]). **Answer:** true

Query: mysubset4([],[],[],[],[]). **Answer:** true

Query: mysubset4([1],[2],[3],[1,2,3],[1,2,3]). **Answer:** true

Query: mysubset4([2],[2],[2],[2],[2]). **Answer:** true

Query: mysubset4([2],[2],[2],[2],[3]). **Answer:** false

- (b) mylisttonum(L,X) returns true iff X is the number represented by the digits in the list L. For example, mylisttonum([2,3,0,1],2301) returns true and mylisttonum([4,3],0) returns false. You are allowed to use in-built arithmetic operators but any other auxiliary predicates has to be defined. **Query:** mylisttonum([1,2,3],X). **Answer:** X=123

Query: mylisttonum([0],X). **Answer:** X=0

Query: mylisttonum([7],X). **Answer:** X=7

Query: mylisttonum([2,3,0,4],X). **Answer:** X=2304

Query: mylisttonum([2,5],X). **Answer:** X=25

Query: mylisttonum([0,2,3,4],X). **Answer:** X=234

(2 Marks)

- (c) myfib(X,Y) returns true iff X is the Yth element in the Fibonacci sequence (0,1,1,2,3,...). For example myfib(0,1), myfib(1,2), myfib(1,3), myfib(2,4), myfib(3,5), myfib(5,6) all return true. (2 Marks)

Query: myfib(X,1). **Answer:** X=0

Query: myfib(X,2). **Answer:** X=1

Query: myfib(X,9). **Answer:** X=21

Query: myfib(X,8). Answer: X=13

Query: myfib(X,11). Answer: X=55.

- (d) mymerge/3. It will be called with two sorted lists as the first two arguments and a free third argument. Such a call should return the merged list of the first two arguments (that is, it is sorted and contains all the elements of the first two arguments). You can assume there are no duplicate elements in the first two lists. For example, mymerge([1,3,5,7],[2,4,10],X) returns X=[1,2,3,4,5,7,10].

(2 Marks)

mymerge([4,6],[1,2,3],[1,2,3,4,5,6]). Answer: true.

mymerge([1,5],[2,4,6],[1,2,3,4,5,6]). Answer: true.

mymerge([1,3],[],[1,3]). Answer: true.

mymerge([],[],[]). Answer: true.

mymerge([1,3],[],[1,3,4]). Answer: false.